Measures Of Location And Spread

Solutions

Chapter Overview

Solution 2

This is identical to the equivalent chapter in the old S1 module. Some content will be familiar from GCSE (mean of grouped/ungrouped data), but many concepts new (e.g. standard deviation) along with possibly unfamiliar notation.

1: Mean, Median, Mode

"Calculate the mean of this grouped frequency table."

2: Quartiles, Percentiles, Deciles

"Use linear interpolation to estimate the interquartile range."

3: Variance & Standard Deviation

"Calculate the standard deviation of the maths marks."

4: Coding

"The marks x were coded using $y = 2x + 10$. Given that the standard deviation of y is 5, determine the standard deviation of x."

Variables in algebra vs stats

Solution 3

$$x$$

Similarities

- Just like in algebra, variables in stats **represent the value of some quantity**, e.g. shoe size, height, colour.
- As we saw in the previous chapter, variables can be discrete or continuous.
- **Can be part of further calculations**, e.g. if x represents height, then $2x$ represents twice people's height. In stats this is known as '**coding**', which we'll cover later.

Differences

- Unlike algebra, a variable in stats represents the value of **multiple objects** (i.e. it's a bit like a set). e.g. the heights of **all** people in a room.
- Because of this, we can do **operations** on it as if it was a **collection of values**:
- If x represents people's heights,

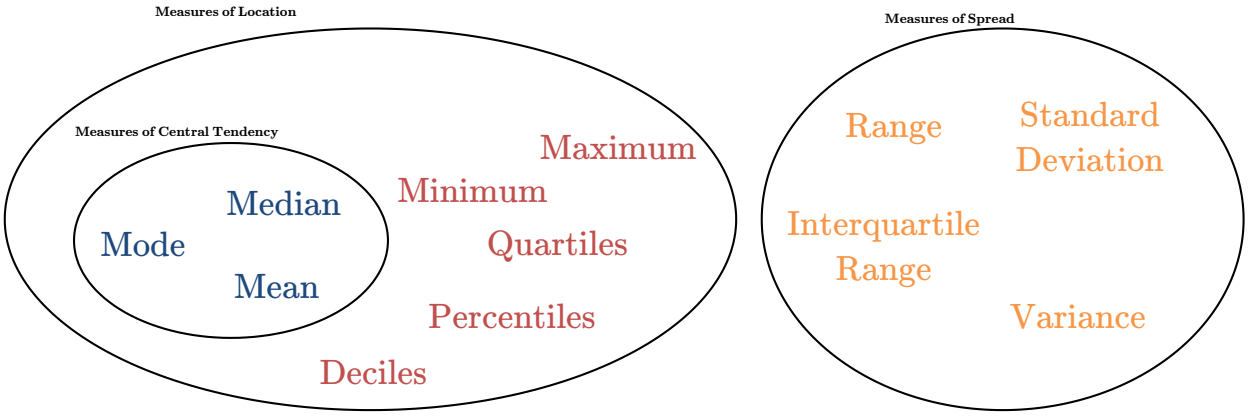
$$\Sigma x$$

gives the sum of everyone's heights. In algebra this would be meaningless: if $x = 4$, then Σx makes no sense!

- $\overline{x}$ is the mean of x. Notice x is a collection of values whereas $\overline{x}$ is a single value.
- To each value of the variable, **we could attach an associated probability**. This is known as a **random variable** (Chapter 6).

Measure of..

Solution 4



Measures of location are single values which describe a **position** in a data set.

Of these, **measures of central tendency** are to do with the **centre of the data**, i.e. a notion of 'average'.

Measures of spread are to do with **how data is spread out**.

Mean of Ungrouped Data

Solution 5

x	162	178	150	160	160	170

You all know how to find the mean of a list of values. But let's consider the notation and see how theoretically we could calculate each of the individual components on a calculator.

$$\bar{x} = \frac{\sum x}{n}$$

The 'overbar' in stats specifically means the sample mean of x

Mean and $\sum x$ on a fx-991CW

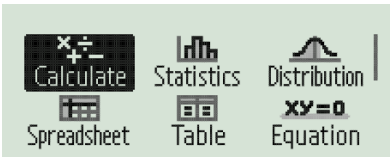
Solution 6

x	162	178	150	160	160	170

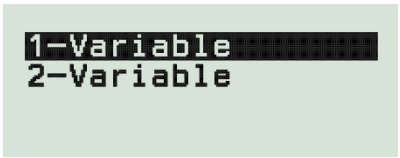
These are instructions for the **Casio fx-570/991CW**



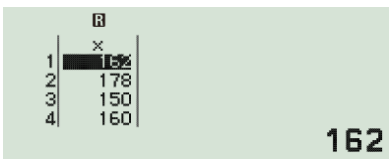
1. Use the arrows and OK to select Statistics.



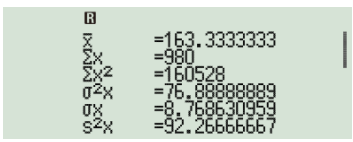
2. Choose 1-Variable.



3. Enter each value with EXE. After the last entry press OK to open the menu.



4. Select **1-Var Results** and read off the values of Σx and n and $\bar{x}$.



Mean and $\sum x$ on a fx-CG50

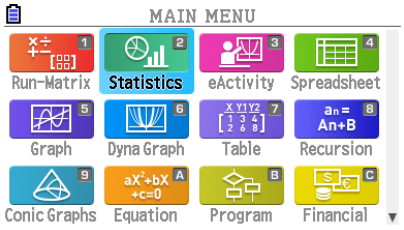
Solution 7

x	162	178	150	160	160	170

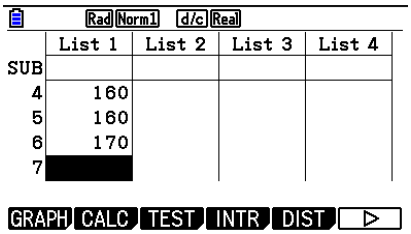
These are instructions for the **Casio fx-CG50**



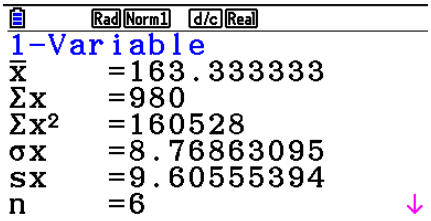
1. Choose 2 for Statistics.



2. Enter the values, pressing EXE after each.



3. Press CALC (F2) then 1-VAR (F1). Read off the values $\overline{x}$ and Σx.



Mean and $\sum x$ on a fx-CG100

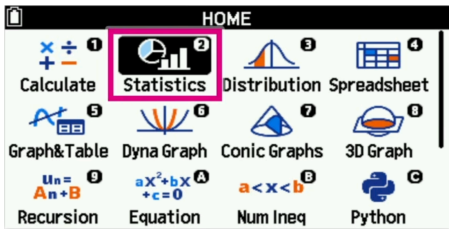
Solution 8

x	162	178	150	160	160	170

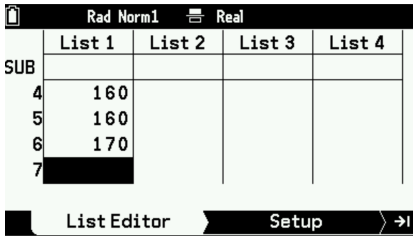
These are instructions for the **Casio fx-CG100**



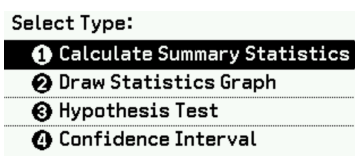
1. Choose 2 for Statistics.



2. Enter the values, pressing EXE after each.

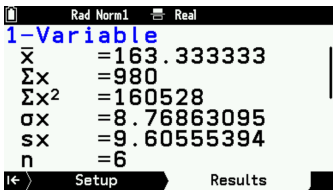


3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and 1 for Freq. Select Execute.





4. Read off the values $\overline{x}$ and Σx.



Now Try this — Solution 9

Diameter of coin **x** (cm)	2.2	2.5	2.6	2.65	2.9

$n =$
5

$\sum x/n =$
2.57

$3\bar{x} + 1 =$
8.71

$\sum x =$
12.85

$\bar{x} =$
2.57

Exercise 2A –Statistics Year 1 page 23 — Solution 10

7. A naturalist records the number of eggs laid by a kākāpō in each brood cycle.

Eggs	1	2	3
Frequency	7	p	2

The mean is 1.5 eggs. Find the missing frequency.

7

$\frac{7+2p+6}{7+p+2} = 1.5$

$13 + 2p = 13.5 + 1.5p \quad \Rightarrow \quad p = 1$

Mean for Ungrouped Frequency Tables Using Σ

Solution 11

The number of children x in different families is recorded.

Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

Determine the mean number of children per family, $\overline{x}$.

Since the frequency tells us how many times each value occurs, the data when listed out in full would be as follows:

0 0 0 0 0 0 1 1 1 1
1 1 1 1 2 2 2 2 2 2
2 2 2 2 2 2 2 2 3 3
3 3 3 3 3 3 4 4 5

The value of 2 appears 14 times.
Therefore, the total of these 2's is

$$fx = 14 \times 2 = 28$$

Σfx therefore represents the total of all these products, and thus the sum of x.

The number of values is the **total frequency**, i.e. $\mathbf{n = \Sigma f}$
Therefore:

For a grouped frequency table:

$$\overline{x} = \frac{\Sigma fx}{\Sigma f}$$

Example

Solution 12

The age of dogs at an animal shelter is recorded:

Age (x)	1	2	3	4	5
Frequency (f)	3	1	0	6	7

Determine the mean age $\overline{x}$.

$$\Sigma fx = (1 \times 3) + (2 \times 1) + (3 \times 0) + (4 \times 6) + (5 \times 7)$$

$$= 64$$

$$\Sigma f = 17$$

$$\overline{x} = \frac{\Sigma fx}{\Sigma f} = \frac{64}{17} = \mathbf{3.8 \text{ years}}$$

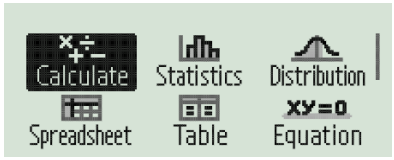
Ungrouped Mean on a fx-991CW

Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

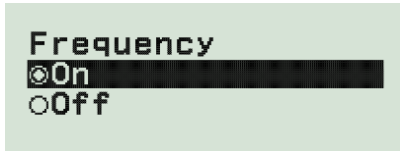
These are instructions for the **Casio fx-570/991CW**



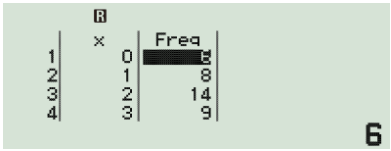
1. Use the arrows and OK to select Statistics. Choose 1-Variable.



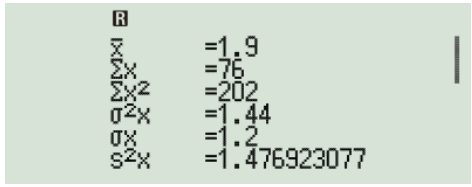
2. Press **Tools** than choose **Frequency**. Choose **On**.



3. Enter the values in x and their frequencies in Freq using EXE. Press OK for results.



4. Select **1-Var Results** and read off the values of $\overline{x}$. Note that **Σx represents Σfx** (since the x in Σx means the original data with duplicate values considered)



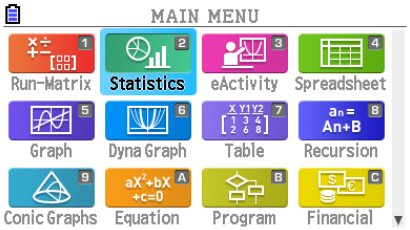
Ungrouped Mean on the fx-CG50

Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

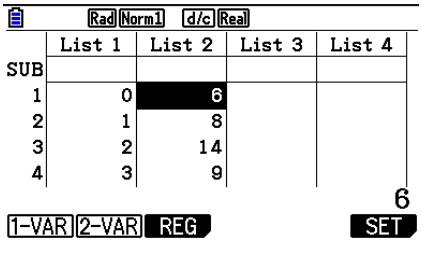
These are instructions for the **Casio fx-CG50**



1. Choose 2 for Statistics.



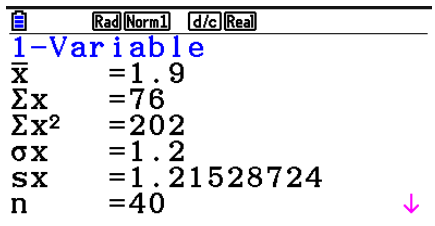
2. Enter the values of x and frequencies in the two columns, pressing EXE after each entry.



3. Press F2 (CALC), then F6 (SET). Set 1Var Freq to List 2 and press EXIT.



Press F1 for 1-Var.



Ungrouped Mean on the fx-CG100

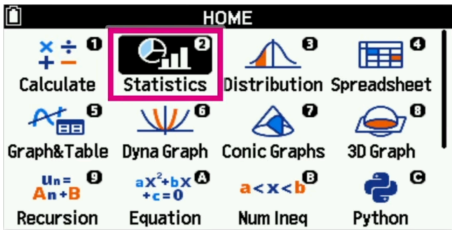
Solution 15

Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

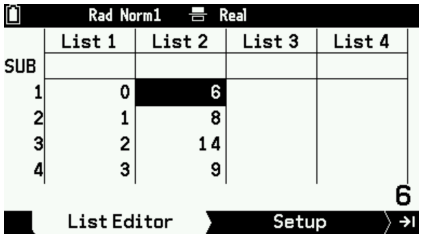
These are instructions for the **Casio fx-CG100**



1. Choose 2 for Statistics.



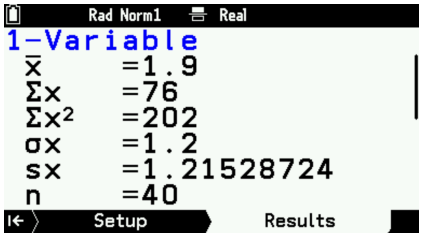
2. Enter the values of x and frequencies in the two columns, pressing EXE after each entry.



3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and List 2 for Freq. Select Execute.



4. Read off the statistics



Test Your Understanding

Solution 16

The age of members of a youth choir is recorded:

Age (x)	10	11	12	13	14	15
Frequency (f)	1	3	4	11	7	4

Determine the mean age $\overline{x}$.

Mean age

$$\Sigma fx = (1 \times 10) + (3 \times 11) + (4 \times 12) + (11 \times 13) + (7 \times 14) + (4 \times 15) = 392$$

$$\overline{x} = \frac{392}{30} = \mathbf{13.1 \text{ years}}$$

Estimating Mean From a Grouped Frequency Table

Solution 17

Example: The weights of 40 randomly caught cats are summarised in the table below.

Weight (kg)	Number of cats, f	Midpoint x
$5-6$	8	**5.5**
$6-8$	12	**7**
$8-10$	15	**9**
$10-12$	5	**11**

Given that x denotes the midpoint of each group in the table and $\mathbf{\Sigma fx = 318}$
calculate an estimate for the mean.

If calculating from scratch:

$\Sigma fx = (8 \times 5.5) + (12 \times 7) + (15 \times 9) + (5 \times 11) = 318$

Then using the normal formula:

$$\mathbf{\overline{x} = \frac{\Sigma fx}{\Sigma f} = \frac{318}{40} = 7.95\ kg}$$

This time the original weights are not known, only the group each appears in.

Our 'best guess' is that each value will be, on average, the midpoint of the range it belongs to (e.g. we assume the 5-6kg cats were each 5.5kg). **We therefore replace each value with this midpoint x.**
In this way, Σfx represents the **estimated** total of the cat's weights, rather than the true (unknown) total.

Test Your Understanding

Solution 18

Actual Exam Question:
The times, in seconds, spent in a queue at a supermarket by 85 randomly selected customers, are summarised in the table below.

Time (seconds)	Number of customers, f
$0-30$	2
$30-60$	10
$60-70$	17
$70-80$	25
$80-100$	25
$100-150$	6

Given that x denotes the midpoint of each group in the table and
$\Sigma fx = 6460 \quad \Sigma fx^2 = 529400$
calculate an estimate for the mean.

(1 mark)

$\frac{6460}{85} = \mathbf{76}$

NOTE: Σfx^2 is only needed if calculating standard deviation, which is covered later.

Further Test Your Understanding — Solution 19

The heights of plants, in centimetres, in a garden centre are given in the table below.

Height (cm)	Number of plants, f
$0 - 10$	7
$10 - 20$	3
$20 - 50$	8
$50 - 100$	12

Given that h denotes the midpoint of each group in the table, calculate an estimate for the mean height $\overline{h}$.

Midpoint h

5
15
35
75

$\Sigma fh = (7 \times 5) + (3 \times 15) + (8 \times 35) + (12 \times 75)$
$= 1260$

$\overline{\mathbf{h}} = \frac{\Sigma \mathbf{fh}}{\Sigma \mathbf{f}} = \frac{\mathbf{1260}}{\mathbf{30}} = \mathbf{42}_{\mathbf{cm}}$

Exercise 2B – Statistics Year 1 page 24-25 — Solution 20

2. Noise levels at 30 locations near a concert were measured to the nearest decibel.

Decibels	65–69	70–74	75–79	80–84	85–89	90–94	95–99
f	1	4	6	6	8	4	1

(a) Estimate the mean noise level.
(b) Explain why it is an estimate.

2a $\bar{x} \approx \frac{2470}{30} = 82.3\,\text{dB}$

2b The class midpoints stand in for the individual readings.

3. Daily mean temperatures at Heathrow in October 1987 are grouped below.

Temperature ($^\circ$C)	6–8	8–10	10–12	12–14	14–16	16–18
f	3	7	9	7	3	2

(a) Give the modal class.
(b) Estimate the mean daily temperature.

3a $10 \leq t < 12$

3b $\bar{t} \approx \frac{353}{31} = 11.4^\circ\text{C}$

4. Two DIY shops recorded the ages of their workers.

Age	16–25	26–35	36–45	46–55	56–65	66–75
A	5	16	14	22	26	14
B	4	12	10	28	25	13

Compare the estimated means. Which shop tends to employ older workers?

4

$\bar{x}_A \approx \frac{4828.5}{97} = 49.8, \qquad \bar{x}_B \approx \frac{4696}{92} = 51.0$

Workers at shop B are about one year older on average.

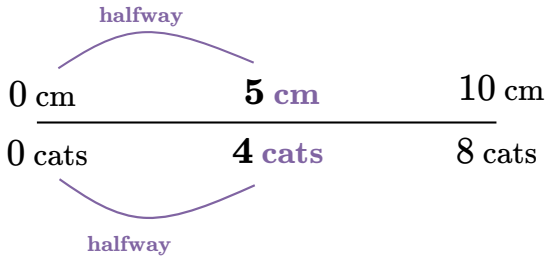
The table shows the heights of various cats.

Height $\mathbf{h}$ (cm)	Frequency
$0 \leq h \leq 10$	8
$10 \leq h \leq 20$	60

Estimate the number of cats with a height:

a. Below 5 cm
b. Above 15 cm
c. Below 12.5 cm

We are assuming the 8 cats are **equally distributed** between 0 and 10 cm, so that there are 4 between $0-5$ cm and 4 between $5-10$ cm.

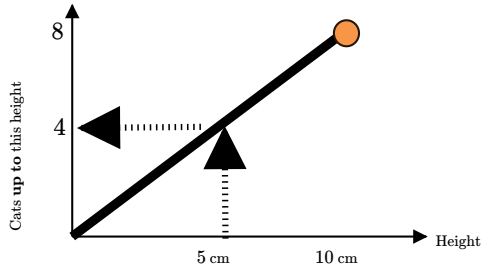


This is known as **linear interpolation**, because if we were plotting this as a cumulative frequency graph (i.e. the running total of cats up to a specific height), **the graph forms a straight line**.

a 4

b 30

c $8 + 15 = 23$



Test Your Understanding

Solution 23

The grouped frequency table shows the times taken for students to complete a puzzle.

Time **t** (secs)	Frequency
$0 \leq t < 20$	12
$20 \leq h < 40$	8
$40 \leq h < 50$	20

Estimate the number of students with a time above 25 seconds.

25 seconds is $\frac{1}{4}$ of the way across the $20 - 40$ interval.
Therefore, include:

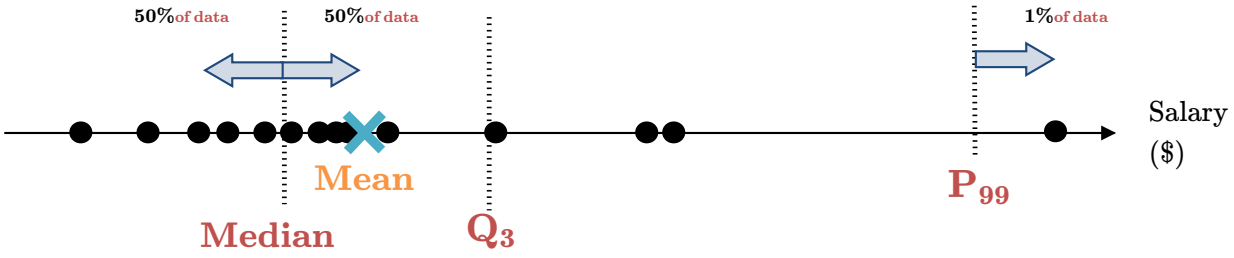
- $\frac{3}{4}$ of 8 students $= 6$
- All of the 20 students

$$\mathbf{6 + 20 = 26}$$

Mean, Median and Measures of Position

Solution 24

MEAN: *Average* that considers all values, but is skewed by extreme values.



MEDIAN: a measure of position because it allows us to get the value a certain position, in this case 50%,along the data.

QUARTILE: gives us the value 25%, 50%, and 75%along the data.
$Q_1 = 25\%$ along data (lower quartile)
$Q_3 = 75\%$ along data (upper quartile)

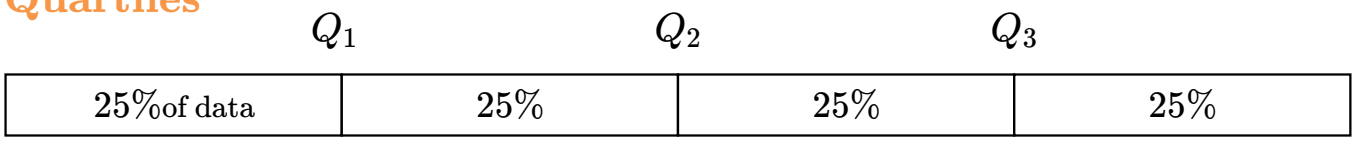
PERCENTILE: % value along the data. For example, the 99th percentile (P_{99}) is the value 99%along the data. When 'the 1%'is used in the media, it's referring to people in the 'top percentile' for salary, i.e. above P_{99}.

Names for Quantiles

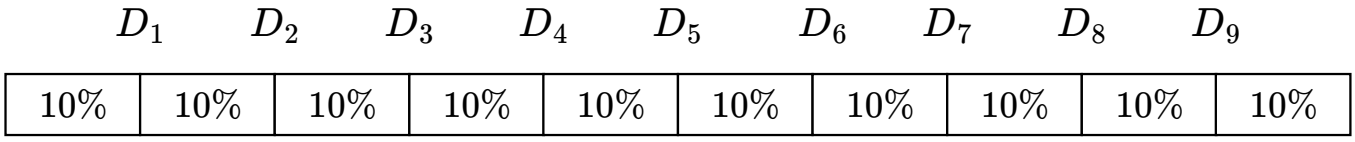
Solution 25

Quantiles represent general positions across the data when ordered.

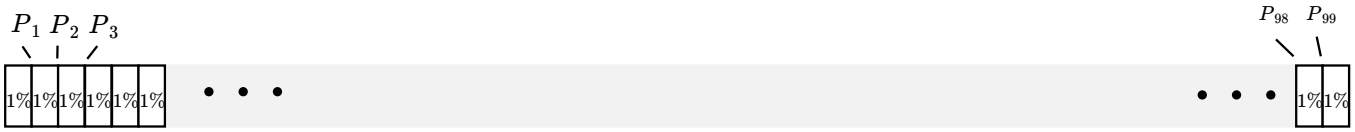
Quartiles



Deciles



Percentiles



Which Item To Use For Listed Data?

Solution 26

Items	n	Position of median	Median
$1, 4, 7, 9, 10$	5	3rd	7
$4, 9, 10, 15$	4	$2^{\text{nd}}/3^{\text{rd}}$	9.5
$2, 4, 5, 7, 8, 9, 11$	7	4th	7
$1, 2, 3, 5, 6, 9, 9, 10, 11, 12$	10	$5^{\text{th}}/6^{\text{th}}$	7.5

Can you think of a rule to find the position of the median given the number of values, n?

The median is the $\frac{1}{2}(n+1)$ th item.

If the position is a decimal, use interpolation.

e.g. the '2.5^{th}' item is halfway between the 2nd and 3rd.

The '4.25^{th}' item is a quarter of the way between the 4th and 5th.

Which Item To Use For Grouped Data?

Solution 27

IQ of L6Ms2 ($\mathbf{q}$)	Frequency ($\mathbf{f}$)
$80 \leq q < 90$	7
$90 \leq q < 100$	5
$100 \leq q < 120$	3
$120 \leq q < 200$	2

If the data is grouped, what item do we use for the median?

median?
$\frac{17}{2} = 8.5^{\text{th}}$ item

To estimate the median of grouped data, calculate $\frac{n}{2}$, then use linear interpolation.

Notice we use $\frac{n}{2}$ rather than $\frac{n+1}{2}$ if grouped data.

Quickfire Questions

Solution 28

What position do we use for the median?

1. Lengths: 3cm, 5cm, 6cm, ...
$n = 11$

Median position: [1] 6th

2. Lengths: 4m, 8m, 12.4m, ...
$n = 24$

Median position: [2] $12^{\text{th}}/13^{\text{th}}$

3. Age	Freq
$10 \leq a < 20$	12
$20 \leq a < 30$	5

Median position: [3] 8.5

4. Score	Freq
$150 \leq s < 200$	3
$200 \leq s < 400$	7

Median position: [4] 5

5. Ages: 5, 7, 7, 8, 9, 10, ...
$n = 60$

Median position: [5] $30^{\text{th}}/31^{\text{st}}$

6. Score	Freq
$150 \leq s < 200$	15
$200 \leq s < 400$	6

Median position: [6] 10.5

7. Weights: 1.2kg, 3.3kg, ...
$n = 35$

Median position: [7] 18th

8. Volume (ml)	Freq
$0 \leq v < 100$	5
$100 \leq v < 200$	6
$200 \leq v < 300$	2

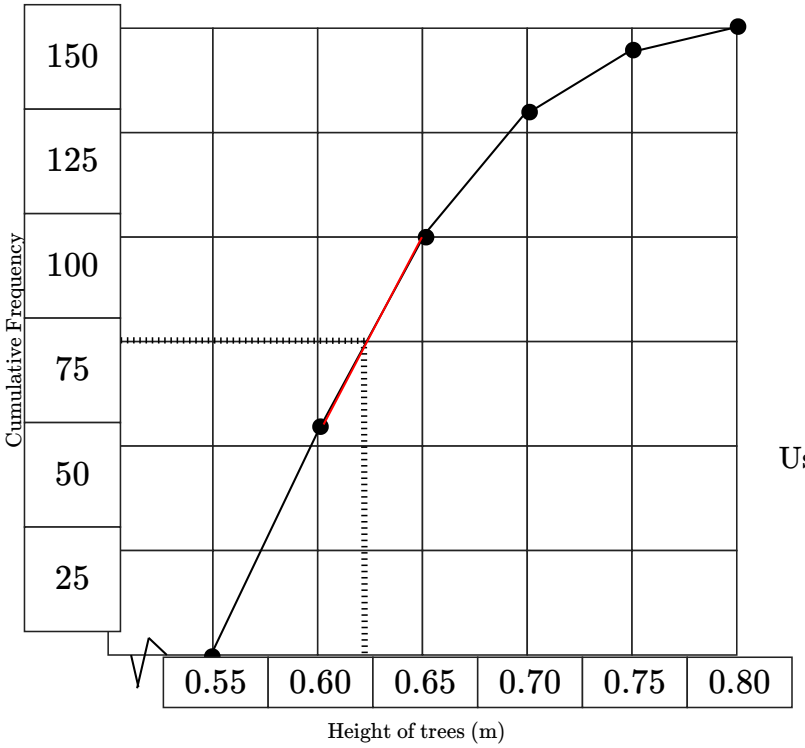
Median position: [8] 6.5

9. Weights: 4.4kg, 7.6kg, 7.7kg...
$n = 18$

Median position: [9] $9^{\text{th}}/10^{\text{th}}$

Linear Interpolation

Solution 29



Height of tree (m)	Freq	C.F
$0.55 \leq h < 0.6$	55	55
$0.6 \leq h < 0.65$	45	100
$0.65 \leq h < 0.7$	30	130
$0.7 \leq h < 0.75$	15	145
$0.75 \leq h < 0.8$	5	150

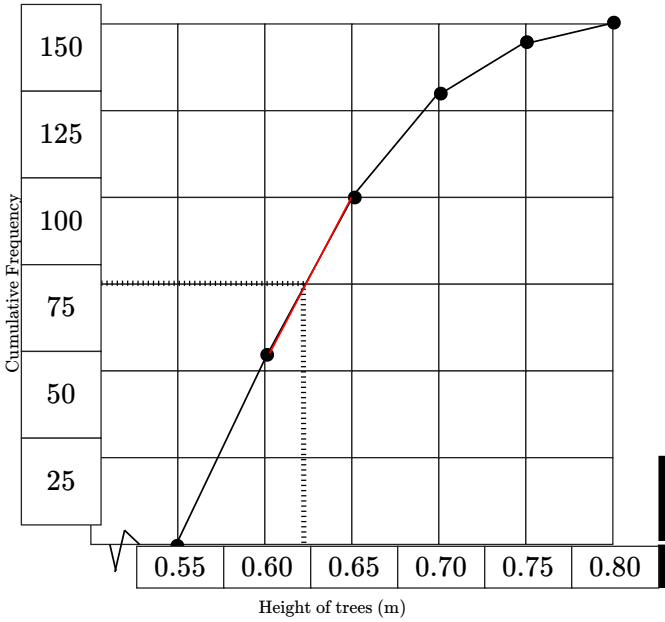
Using a cumulative frequency graph, we know we can estimate the median by drawing a **suitable line**.

How could we read off this value exactly using a suitable calculation?

We could find the fraction of the way along the line segment using the frequencies, then go this same fraction along the class interval.

Linear Interpolation

Solution 30



Height of tree (m)	Freq	C.F
$0.55 \leq h < 0.6$	55	55
$0.6 \leq h < 0.65$	45	100
$0.65 \leq h < 0.7$	30	130
$0.7 \leq h < 0.75$	15	145
$0.75 \leq h < 0.8$	5	150

Use linear interpolation to estimate the median.

Step 1: Identify the **interval** in which the median item, here the $\frac{150}{2} = 75^{\text{th}}$ value, lies.

Step 2: Write the relevant data for the calculation

Frequency up until this interval 55	Item number we're interested in? 75	Frequency by end of this interval? 100
Height at start of interval? 0.6 m	Quartile? Q_2	Height by end of interval? 0.65 m

Linear Interpolation

Solution 31

Height of tree (m)	Freq	C.F.
$0.55 \leq h < 0.6$	55	55
$0.6 \leq h < 0.65$	45	100
$0.65 \leq h < 0.7$	30	130
$0.7 \leq h < 0.75$	15	145
$0.75 \leq h < 0.8$	5	150

What fraction of the way across the interval are we?

$\frac{20}{45}$

Hence:

Median = $0.6 + \frac{20}{45} \times 0.05$

Frequency up until this interval → 55

Item number we're interested in. ← 75

Frequency by end of this interval ← 100

0.6m — Med — 0.65m

Height at start of interval.

Height by end of interval.

Tip: put the units to avoid getting frequencies confused with values of the variable.

Tip: To quickly get frequency before and after, just look for the two cumulative frequencies that surround the item number.

More Examples

Solution 32

1. Weight of cat (kg)	Freq	C.F.
$1.5 \leq w < 3$	10	10
$3 \leq w < 4$	8	18
$4 \leq w < 6$	14	32

Median class interval: $\mathbf{3 \leq w < 4}$

10 — 16 — 18

3kg — Med — 4kg

Fraction along interval: $\frac{6}{8}$

Median: $\mathbf{3 + \left(\frac{6}{8} \times 1\right) = 3.75kg}$

2. Time (s)	Freq	C.F.
$8 \leq t < 10$	4	4
$10 \leq t < 12$	3	7
$12 \leq t < 14$	13	20

7 — 10 — 20

12s — Med — 14s

Median:

$\mathbf{12 + \left(\frac{3}{13} \times 2\right) = 12.5}$ **(to 3sf)**

What's different about the interval here?

Solution 33

Weight of cat to nearest kg	Frequency
$10 - 12$	7
$13 - 15$	2
$16 - 18$	9
$19 - 20$	4

There are **GAPS** between intervals!
What interval does this **actually** represent?

$$10 - 12$$

Lower class boundary? 9.5 – upper class boundary? 12.5

Class Width? 3

Identify the class width

Solution 34

1. Distance **d** travelled (in m)	...
$0 \leq d < 150$	
$150 \leq d < 200$	
$\mathbf{200 \leq d < 210}$	

Lower class boundary = 200

Class width = 10

3. Time **t** taken (in seconds)	...
$0 - 3$	
$\mathbf{4 - 6}$	
$7 - 11$	

Lower class boundary = 3.5

Class width = 3

2. Weight **w** in kg	...
$10 - 20$	
$21 - 30$	
$\mathbf{31 - 40}$	

Lower class boundary = 30.5

Class width = 10

4. Speed **s** (in mph)	...
$10 \leq s < 20$	
$20 \leq s < 29$	
$\mathbf{29 \leq s < 31}$	

Lower class boundary = 29

Class width = 2

Linear Interpolation with gaps

Solution 35

Example:

Summarised below are the distances, to the nearest mile, travelled to work by a random sample of 120 commuters.

Use linear interpolation to estimate the distribution's median

Distance (nearest mile)	Commuters	
0–9	10	10
10–19	19	29
20–29	43	72
30–39	25	97
40–49	8	105
50–59	6	111
60–69	5	116
70–79	3	119
80–89	1	120

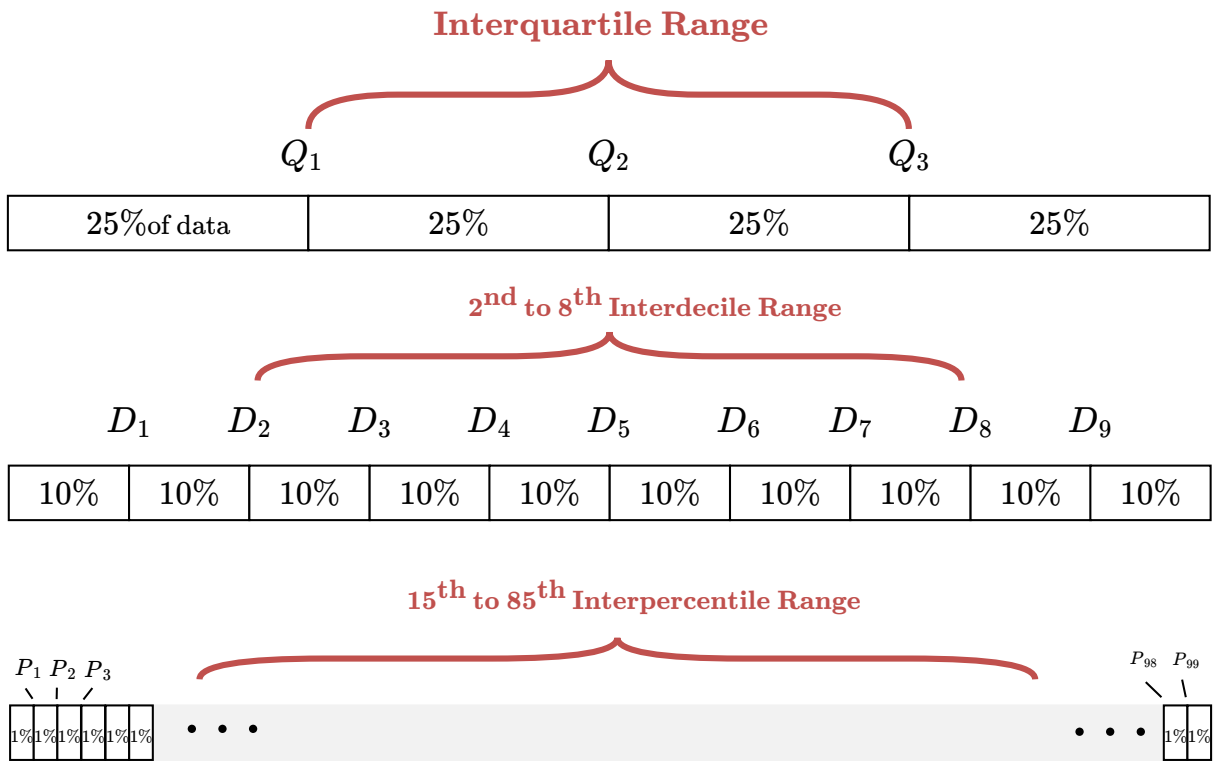
29 | 60 | 72

19.5 miles | Med | 29.5 miles

Median $= 19.5 + \frac{31}{43} \times 10$
$= 26.7$ miles

Inter-percentile and Inter-decile Ranges

Solution 36



Just as we can calculate the interquartile range to mean the range of the middle 50% of data, we can also use deciles and percentiles to find the range of a more general middle percentage of the data.

Test Your Understanding — Solution 37

Use linear interpolation to estimate the 3rd to 7th interdecile range of the following data:

Mass (x) to the nearest gram	Freq ($\mathbf{f}$)
$10 - 12$	5
$13 - 14$	15
$15 - 18$	7
$19 - 22$	3

C.F.: 5, 20, 27, 30

30 items

$D_3 = \frac{3}{10} \times 30 = 9^{\text{th}}$ item

$D_7 = \frac{7}{10} \times 30 = 21^{\text{st}}$ item

5	9	20
12.5 g	D_3	14.5g

$D_3 = 12.5 + \left(\frac{4}{15} \times 2\right) = \mathbf{13.03\ g}$

20	21	27
14.5 g	D_7	18.5g

$D_7 = 14.5 + \left(\frac{1}{7} \times 4\right) = \mathbf{15.07\ g}$

$\mathbf{IDR_{3,7} = 2.04\ g\ to\ 3sf}$

Exercise 2C Page 27-28 — Solution 38

3. A hotel records the number of lift breakdowns each week for 26 weeks.

Breakdowns	0–1	2–3	4–5
f	18	7	1

Use interpolation to estimate the median number of breakdowns. (2 marks)

3

$n/2 = 13, \qquad m \approx 0 + \frac{13}{18}(1.5) = 1.08$

This estimate treats a discrete count as continuous within the first group.

Supplementary exercises

Solution 39

1. Estimate the median daily attendance at a hospital trauma clinic.

Patients	10–19	20–29	30–34	35–39	40–44	45–49	50–69
f	2	18	24	30	27	14	5

1

$34.5 + \frac{60-44}{30} \times 5 \approx 37.2$

2. Use interpolation to estimate the median age of the 300 houses.

Age (years)	0–20	20–40	40–60	60–100	100–200	200–300	300–500
f	36	92	74	39	14	27	18

2

$40 + \frac{150-128}{74} \times 20 \approx 45.9$ years

3. Estimate the median length of a visit to the cyber-café.

Stay (min)	0–30	30–60	60–90	90–120	120–240	240–360
f	15	31	32	23	17	2

3

$60 + \frac{60-46}{32} \times 30 \approx 73.1$ minutes

Supplementary exercises

Solution 40

4. The following table summarises the times, t minutes to the nearest minute, recorded for a group of students to complete an exam.

Minutes	11–20	21–25	26–30	31–35	36–45	46–60
f	62	88	16	13	11	10

You may use $\sum ft^2 = 134281.25$.

a) Estimate the mean of this data

b) Use linear Interpolation to estimate the value of the median

a $\textbf{Mean} = \frac{4837.5}{200} = \textbf{24.1875}$

b

$\textbf{Median} = 20.5 + \left(\frac{38}{88} \times 5\right) = 22.7$

Quartile – which items

Solution 41

You need to be able to find the **quartiles** of both listed data and of grouped data. The rule is exactly the same as for the median.

Listed data

Items	**n**	Position of LQ & UQ	LQ & UQ
$1, 4, 7, 9, 10$	5	2nd & 4th	4 & 9
$4, 9, 10, 15$	4	$1^{\text{st}}/2^{\text{nd}}$ & $3^{\text{rd}}/4^{\text{th}}$	6.5 & 12.5
$2, 4, 5, 7, 8, 9, 11$	7	2nd & 6th	4 & 9
$1, 2, 3, 5, 6, 9, 9, 10, 11, 12$	10	3rd and 8th	3 & 10

Can you think of a rule to find the position of the LQ/UQ given n?

To find the position of the LQ/UQ for listed data, find $\frac{1}{4}n$ or $\frac{3}{4}n$ then as before:
- If a decimal, round up.
- If whole, use halfway between this item and the one after.

Grouped data

IQ of L6Ms2 (**q**)	Frequency (**f**)
$80 \leq q < 90$	7
$90 \leq q < 100$	5
$100 \leq q < 120$	3
$120 \leq q < 200$	2

Position to use for LQ:

4.25

To find the LQ and UQ of grouped data, find $\frac{1}{4}n$ and $\frac{3}{4}n$, then use linear interpolation.

Again, **DO NOT** round this value.

Percentiles

Solution 42

The LQ, median and UQ give you 25%, 50% and 75% along the data respectively.

$n = 43$

Item to use for 57th percentile?



$43 \times 0.57 = 24.51$

You will always find these for grouped data in an exam, **so never round this position**.

Notation:

Lower Quartile: Q_1

Median: Q_2

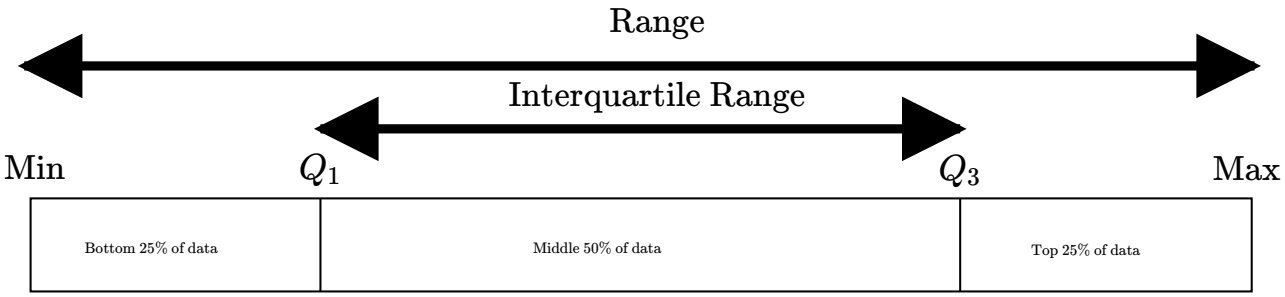
Upper Quartile: Q_3

57th Percentile: P_{57}

Measure of spread

Solution 43

The interquartile range and interpercentile range are examples of **measures of spread.**



Interquartile Range = Upper Quartile − Lower Quartile

Why might we favour the interquartile range over the range?

Because it gives us the spread of the data excluding the extreme values at either end.

We can control this further by having for example the "*10th to 90th interpercentile range*", which would be $P_{90} - P_{10}$. This is "*the range of the data with the most extreme 10% of values at either end excluded*".

The 10th percentile is also known as the 1st **decile** (D_1), and similarly $P_{90} = D_9$.

Test your understanding

Solution 44

These are the same as the 'Test Your Understanding' questions on your supplementary sheet from before.

Age of relic (years)	Frequency
0-1000	24
1001-1500	29
1501-1700	12
1701-2000	35

Item for Q_1:

25th

$Q_1 = 1000.5 + \left(\frac{1}{29} \times 500\right)$
$= 1017.74$ years

$Q_3 = 1700.5 + \left(\frac{10}{35} \times 300\right)$
$= 1786.21$ years

$IQR = 768.47$

Shark length (cm)	Frequency
$40 \leq x < 100$	17
$100 \leq x < 300$	5
$300 \leq x < 600$	8
$600 \leq x < 1000$	11

Item to use for P_{10}:

$41 \times 0.1 = 4.1$

$P_{10} = \mathbf{40} + \left(\frac{\mathbf{4.1}}{\mathbf{17}} \times \mathbf{60}\right) = \mathbf{54.47}$

$P_{90} = \mathbf{600} + \left(\frac{\mathbf{6.9}}{\mathbf{11}} \times \mathbf{400}\right) = \mathbf{850.91}$

10th to 90th interpercentile range

$850.91 - 54.47 = 796.44$

5. A roadside assistance firm recorded customer waiting times.

Minutes	20–30	30–40	40–50	50–60	60–70
f	6	10	18	13	2

(a) Estimate the mean waiting time. (1 mark)

(b) Calculate the 65th percentile. (2 marks)

(c) Only 10% of customers wait longer than 56 minutes, the firm claims. Calculate a suitable percentile and comment. (3 marks)

5a $\bar{t} \approx 2155/49 = 44.0$ minutes

5b $P_{65} \approx 40 + 10(31.85 - 16)/18 = 48.8$

5c

$P_{90} \approx 50 + 10(44.1 - 34)/13 = 57.8$

The estimated 90th percentile exceeds 56 minutes, so the claim is unsupported.

6. The wingspans of 100 Californian condors are recorded in metres.

Wingspan	1–1.5	1.5–2	2–2.5	2.5–3	3+
f	4	20	37	28	11

(a) Estimate and interpret the 80th percentile. (3 marks)

(b) Explain why interpolation cannot estimate the 90th percentile. (1 mark)

6a

$P_{80} \approx 2.5 + \frac{80-61}{28}(0.5) = 2.84\,\text{m}$

About 80% have wingspans below 2.84 m.

6b

The final class contains the 90th percentile and has no upper boundary.

2. Monthly incomes of factory workers are grouped in pounds.

Income (£)	900–1000	1000–1100	1100–1200	1200–1300
f	3	24	28	15

(a) Estimate the 34th to 66th interpercentile range. (3 marks)

(b) Estimate the number of incomes within this range. (2 marks)

2a

$P_{34} \approx 1000 + 100(23.8 - 3)/24 = 1086.67$

$P_{66} \approx 1100 + 100(46.2 - 27)/28 = 1168.57$

$P_{66} - P_{34} \approx £81.90$

2b $0.32(70) \approx 22$ workers

3. Lancaster to Preston journey times were recorded to the nearest minute.

Minutes	15–16	17–18	19–20	21–22
f	5	10	35	10

(a) Estimate the 5th to 95th interpercentile range. (3 marks)

(b) Estimate the number of journeys within this range. (1 mark)

3a

$P_5 = 14.5 + 2(3/5) = 15.7, \quad P_{95} = 20.5 + 2(7/10) = 21.9$

$21.9 - 15.7 = 6.2$ minutes

3b

$0.90(60) = 54$ journeys

4. Daily mean temperatures in Leeming for the first ten days of June 1987 (°C):
14.3, 12.7, 12.4, 10.9, 9.4, 13.2, 12.1, 10.3, 10.3, 10.6
(a) Calculate the median and IQR. (2 marks)
(b) Compare June with May (median 9.9°C, IQR 3.9°C). (2 marks)
(c) In July the 10th to 90th interpercentile range was 5.4°C. Estimate how many days fell in it. (1 mark)

4a
$Q_2 = (10.9 + 12.1)/2 = 11.5, \quad Q_3 - Q_1 = 12.7 - 10.3 = 2.4$

4b
June had a higher typical temperature and a smaller spread.

4c
$(0.90 - 0.10) \times 31 = 24.8 \approx 25$ days

1. Exam completion times were recorded to the nearest minute.

Minutes	11–20	21–25	26–30	31–35	36–45	46–60
f	62	88	16	13	11	10

(c) Show that the lower quartile is 18.6 minutes. (d) Estimate the IQR.

1
$Q_1 = 10.5 + 10(50/62) \approx 18.6, \quad Q_3 = 25.5$
$\text{IQR} \approx 25.5 - 18.6 = 6.9$ min

2. Distances travelled to London by 134 examiners, to the nearest km:

Distance (km)	41–45	46–50	51–60	61–70	71–90	91–150
f	4	19	53	37	15	6

Use interpolation to estimate the median and both quartiles.

2
$Q_1 = 50.5 + 10(33.5 - 23)/53 \approx 52.5$
$Q_2 \approx 58.8$ km, $\quad Q_3 = 60.5 + 10(100.5 - 76)/37 \approx 67.1$ km

3. Estimate both quartiles and the IQR for the ages of 300 houses.

Age (years)	0–20	20–40	40–60	60–100	100–200	200–300	300–500
f	36	92	74	39	14	27	18

3

$Q_1 = 20 + 20(75 - 36)/92 \approx 28.48, \quad Q_3 = 60 + 40(225 - 202)/39 \approx 83.59$

IQR ≈ 55.11 years

4. Estimate both quartiles and the 90th percentile for café visits.

Stay (min)	0–30	30–60	60–90	90–120	120–240	240–360
f	15	31	32	23	17	2

4

$Q_1 \approx 44.52, \quad Q_3 \approx 105.65, \quad P_{90} \approx 169.41$ minutes

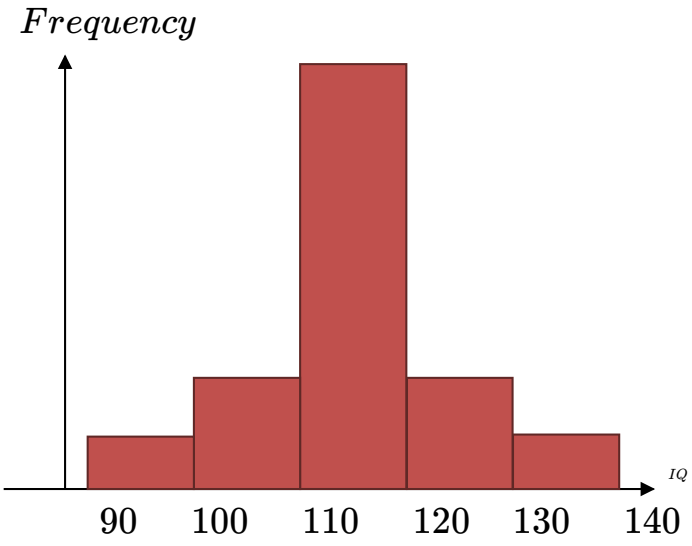
5. Commuter distances, to the nearest mile. Estimate the IQR.

Miles	0–9	10–19	20–29	30–39	40–49	50–59	60–69	70–79	80–89
f	10	19	43	25	8	6	5	3	1

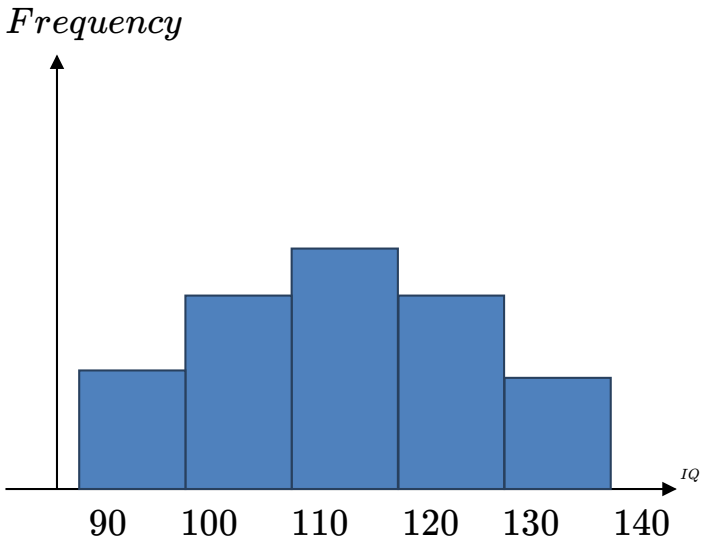
5

$Q_1 = 19.5 + 10/43, \quad Q_3 = 29.5 + 10(90 - 72)/25, \quad$ IQR ≈ 16.97 miles

Distribution of IQs in Class A



Distribution of IQs in Class B



Here are the distribution of IQs in two classes of the same size. What's the same, and what's different?

The (estimated) mean IQ is the same for the two classes, as is the (estimated) range, but the overall spread of values is greater for the second class. The interquartile range would convey this, but do we have a method of measuring spread that takes into account all the values?

Variance

Solution 53

Variance is a measure of spread that takes all values into account.
Variance, by definition, is the **average squared distance from the mean.**

$$\sigma^2 = \frac{\sum(x-\bar{x})^2}{n}$$

Notation: While Σ is 'uppercase sigma' and means 'sum of', σ is 'lowercase sigma' (we'll see why we have the squared in a sec)

Distance from mean. . .

Squared distance from mean. . .

Average squared distance from mean. . .

Simpler formula for variance

Solution 54

In practice, it is possible to simplify the formula to the following*:

Variance mean of the squares minus the square of the mean:

$$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$$

Proof (outside the syllabus)

*** Proof: (not in syllabus)**

Note that $\overline{x}$ is constant for a fixed variable, and that in general, $\Sigma k f(x) = k\Sigma f(x)$ for a constant k, i.e. we can factor out constants out of a summation.

$$\sigma^2 = \frac{\Sigma(x-\overline{x})^2}{n} = \frac{\Sigma\left(x^2-2x\overline{x}+\overline{x}^2\right)}{n}$$
$$= \frac{\Sigma x^2}{n} - \frac{\Sigma(2x\overline{x})}{n} + \frac{\Sigma\overline{x}^2}{n}$$
$$= \frac{\Sigma x^2}{n} - 2\overline{x}\left(\frac{\Sigma x}{n}\right) + \frac{\overline{x}^2}{n}\Sigma 1$$
$$= \frac{\Sigma x^2}{n} - 2\overline{x}^2 + \frac{\overline{x}^2}{n} \cdot n$$
$$= \frac{\Sigma x^2}{n} - 2\overline{x}^2 + \overline{x}^2$$
$$= \frac{\Sigma x^2}{n} - \overline{x}^2$$

Standard Deviation

$$\sigma = \sqrt{Variance}$$

The standard deviation **can 'roughly' be thought of as the average distance from the mean.**

Standard Deviation

Solution 55

3, 11

Variance

$\sigma^2 = 65 - 7^2 = 16$

Standard deviation

$\sigma = \sqrt{16} = 4$

Note that in the case of two items, the standard deviation **is** indeed the average distance of the values from the mean.

2cm 3cm 3cm 5cm 7cm

Variance

$\sigma^2 = 19.2 - 4^2 = 3.2\,\text{cm}^2$

Standard deviation

$\sigma = \sqrt{3.2} \approx 1.79\,\text{cm}$

Practice

Solution 56

Find the variance and standard deviation of the following sets of data.

2 4 6

Variance = 2.67

Standard Deviation = 1.63

1 2 3 4 5

Variance = 2

Standard Deviation = 1.41

σ on a fx-991CW

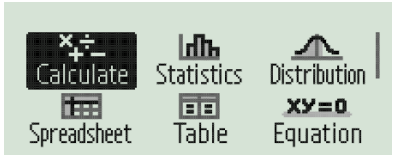
Solution 57

Time (secs)	30	34	35	39

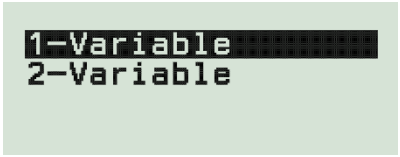
These are instructions for the **Casio fx-570/991CW**



1. Use the arrows and OK to select Statistics.



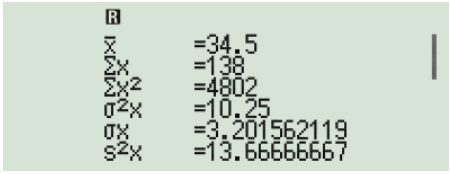
2. Choose 1-Variable.



3. Enter each value with EXE. After the last entry press OK to open the menu.



4. Choose '1 Variable Results'. Read off σx (σ) and $\sigma^2 x$ (the variance σ^2)



σ on a fx-CG50

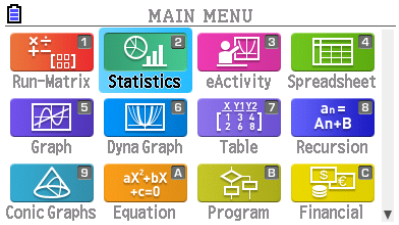
Solution 58

Time (secs)	30	34	35	39

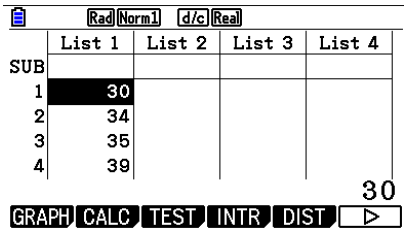
These are instructions for the **Casio fx-CG50**



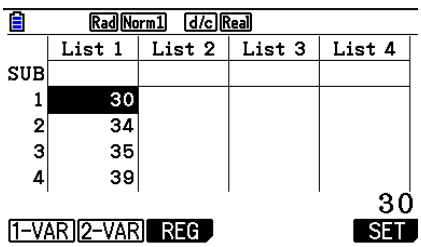
1. Choose 2 for Statistics.



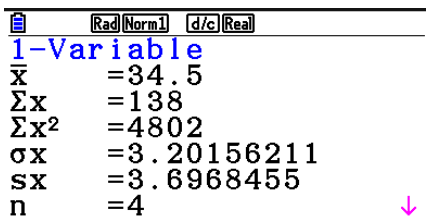
2. Enter your values, pressing EXE after each and using the arrow keys to move around.



3. In CALC (F2), open SET (F6). Set 1Var Freq to 1, press EXIT, then 1-VAR (F1).



4. Read off σ.

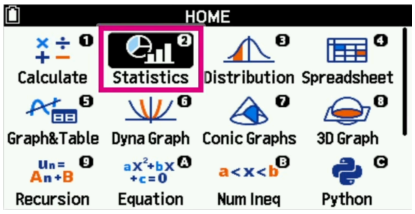


Time (secs)	30	34	35	39

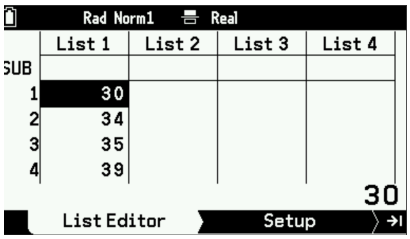
These are instructions for the **Casio fx-CG100**



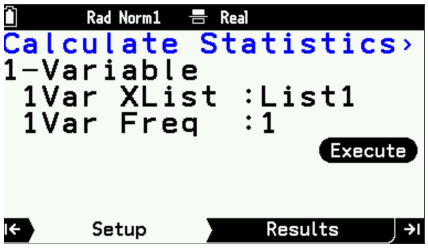
1. Choose 2 for Statistics.



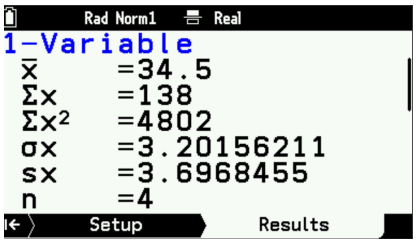
2. Enter your values, pressing EXE after each and using the arrow keys to move around.



3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and 1 for Freq. Select Execute.



4. Read off σ.



Test Your Understanding

Solution 60

Use a calculator to determine the standard deviation of the following data set of long jump distances.

6.1 m, 8.3 m, 8.3 m, 8.4 m, 8.5 m, 8.7 m

Read off directly from your calculator.

$\sigma = \mathbf{0.8827}$ m to 4dp

Calculating σ Using Given Summations

Solution 61

It is unlikely in an exam you would be asked to compute the standard deviation from scratch. Most likely you would calculate σ directly without formula or be given key summations to use in the formula.

EXAMPLE

Before going on holiday to Seapron, Tania records the weekly rainfall (x mm) at Seapron for 8 weeks during the summer. Her results are summarised as

$$\Sigma x = 86.8 \qquad \Sigma x^2 = 985.88$$

$$\sigma^2 = \frac{\Sigma x^2}{n} - \overline{x}^2$$

Find the standard deviation, σ_x, for these data.

(3 marks)

1. Calculate $\overline{x}$ using $\frac{\Sigma x}{n}$:

$\overline{x}$ $\overline{x} = \frac{\Sigma x}{n} = \frac{86.8}{8} = 10.85$

2. Use the formula with provided Σx^2:

σ^2 $\sigma^2 = \frac{985.88}{8} - 10.85^2 = 5.5125$

3. Calculate σ_x :

σ $\sigma = \sqrt{5.5125} = \mathbf{2.35}$ mm to 3sf

Test Your Understanding

Solution 62

[Edexcel AS SAM P2 Q1c]

Sara is investigating the variation in daily maximum gust, t kn, for Camborne in June and July 1987. She used the large data set to select a sample of size 20 from the June and July data for 1987.

Sara selected the first value using a random number from 1 to 4 and then selected every third value after that.

The data Sara collected are summarised as follows

$$\text{n} = 20 \qquad \Sigma\text{t} = 374 \qquad \Sigma\text{t}^2 = 7600$$

Calculate the standard deviation.

(2 marks)

Mark scheme

$\bar{t} = 374/20 = 18.7$

$\sigma_t = \sqrt{7600/20 - \bar{t}^2} = \sqrt{30.31}$ (M1)

$\sigma_t = 5.5054\ldots \approx 5.51$ (A1)

Also accept $s_t = \sqrt{(7600 - 20\bar{t}^2)/19} = 5.6484\ldots$

Extending to frequency/grouped frequency tables

Solution 63

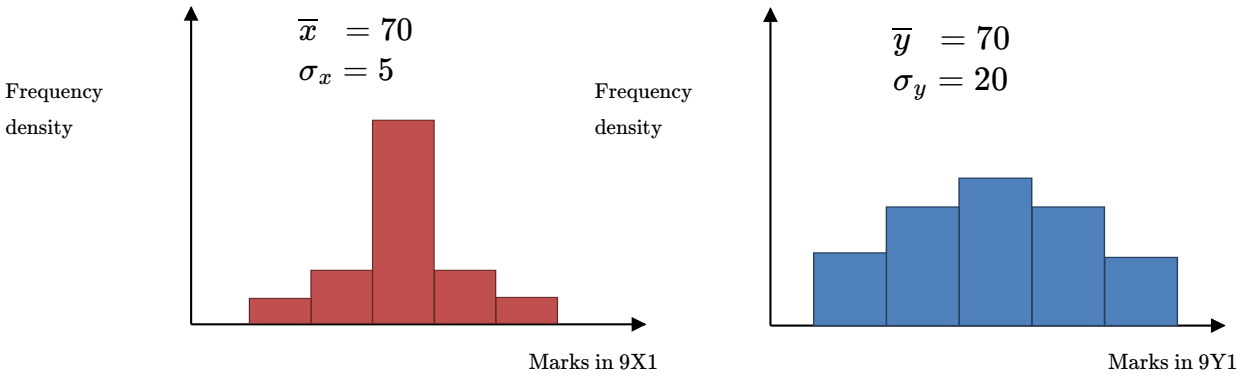
You may want to use the following mnemonic:

Variance: "The mean of the squares minus the square of the mean ('msmsm')"

$$\text{Variance} = \frac{\sum fx^2}{\sum f} - \bar{x}^2$$

Comparing Data Sets Using Standard Deviation

Solution 64



Compare the marks of the two classes.

The classes had the same mean mark, but 9X1 had a lower standard deviation, suggesting the marks were **more consistent**.

Note: When σ is lower, it means there's less variation, which means **more consistency**.

Previously when comparing two data sets, we might compare:

1. A measure of **central tendency** (usually the mean)
2. A measure of **spread**

For the latter, you may have previously used the range or interquartile range. **Now you can use the standard deviation!**

Test Your Understanding — Solution 65

Exam Question

The number of hours of sunshine each day, y, for the month of July at Heathrow are summarised in the table below.

Hours y	$0 \leq y < 5$	$5 \leq y < 8$	$8 \leq y < 11$	$11 \leq y < 12$	$12 \leq y < 14$
f	12	6	8	3	2

It can be shown that the mean of this data is 6.63 and that the standard deviation is 3.69.

The mean and standard deviation for the number of hours of daily sunshine for the same month in Hurn are 5.98 hours and 4.12 hours respectively.

Thomas believes that the further south you are the more consistent should be the number of hours of daily sunshine.

State whether or not the standard deviations for the two locations support Thomas' belief.

Mark Scheme?

Heathrow has the higher mean and the smaller standard deviation, so its sunshine hours are more consistent. Hurn lies south of Heathrow. The data do not support Thomas's belief.

Effect of an Added Value on σ — Solution 66

If we consider the standard deviation as, approximately, the average distance of each value from the mean, we can reflect on how additional values will impact the standard deviation.

The populations of 10 countries are recorded. The mean population is 9 million, and the standard deviation is 1 million.

I add Switzerland to my data set, which has a population of 9 million.

State the effect on:

a. the range of the data

b. the standard deviation of the data

a

As 9 million is the mean of the existing data set, with the minimum value of all the data below 9 million and the maximum above it, the added value **will not affect the range**.

b

Since the deviation of 9 million from the mean is 0, this reduces the mean squared deviation, and therefore the variance (and hence the standard deviation) **will be reduced**.

Standard Deviation From Grouped Data

Solution 67

Data is...	Mean	Variance
Ungrouped	$\overline{x} = \frac{\Sigma x}{n}$	$\sigma^2 = \frac{\Sigma x^2}{n} - \overline{x}^2$
Grouped	$\overline{x} = \frac{\Sigma fx}{\Sigma f}$	$\sigma^2 = \frac{\Sigma fx^2}{\Sigma f} - \overline{x}^2$

MEAN

If the data is grouped, then we previously saw that to calculate the mean, fx gives the total when x is written out f times, so accounts for the duplication of values.

$\Sigma f = n$, because the total frequency is the number of data values.

VARIANCE

Each value is duplicated f times, so each squared value will also be duplicated f times, contributing fx^2 to the total.

Example With Given Summations

Solution 68

EXAM QUESTION

The following table summarises the times, t, minutes to the nearest minute, recorded for a group of students to complete an exam.

Time t (min)	11–20	21–25	26–30	31–35	36–45	46–60
Students f	62	88	16	13	11	10

[You may use $\Sigma ft^2 = 134281.25$*]*

Estimate the mean and standard deviation of these data. **(5)**

Find the class midpoints.

$15.5, 23, 28, 33, 40.5, 53$

Estimate the mean.

$$\overline{t} = \frac{\Sigma ft}{\Sigma f} = \frac{15.5\times62+\ldots+53\times10}{200} = 24.1875$$

Estimate the standard deviation.

$$\sigma^2 = \frac{134281.25}{200} - 24.1875^2$$

$$\sigma = \sqrt{86.37} = \mathbf{9.29}$$

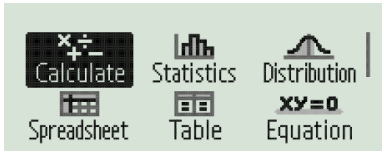
σ on for Grouped Data on a fx-991CW

Time (minutes) t	$11-20$	$21-25$	$26-30$	$31-35$	$36-45$	$46-60$
Midpoints	15.5	23	28	33	40.5	53
Number of students f	62	88	16	13	11	10

These are instructions for the **Casio fx-570/991CW**



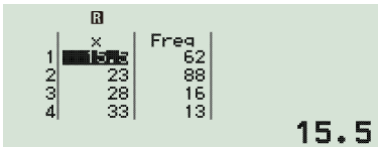
1. Use the arrows and OK to select Statistics.



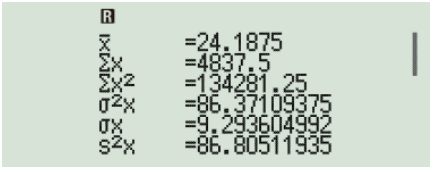
2. Choose 1-Variable. Open Tools, set Frequency to On, then press Back.



3. Enter the values in x and their frequencies in Freq using EXE. Press OK for results.



4. Choose '1 Variable Results'. Read off σx (σ) and $\sigma^2 x$ (the variance σ^2)



Important note: Σx^2 here means $\Sigma f x^2$, consistent with the value given in the question on the previous slide. The frequencies are implicitly 'baked in'.

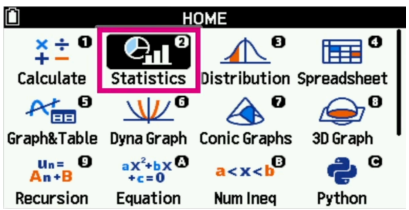
σ on for Grouped Data on a fx-CG100

Time (minutes) t	$11-20$	$21-25$	$26-30$	$31-35$	$36-45$	$46-60$
Midpoints	15.5	23	28	33	40.5	53
Number of students f	62	88	16	13	11	10

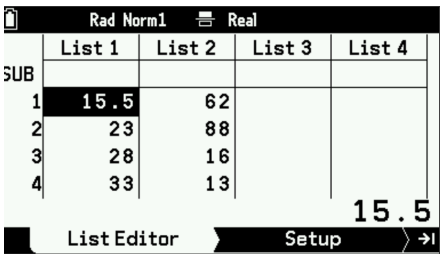
These are instructions for the **Casio fx-CG100**



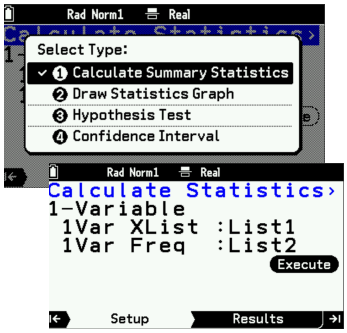
1. Choose 2 for Statistics.



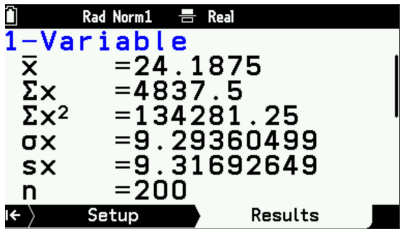
2. Enter the midpoints in List 1 and the Frequencies in List 2



3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and List 2 for Freq. Select Execute.



4.Read off σx (σ). Square this value for the variance.



Test Your Understanding — Solution 71

Exam Question

An agriculturalist is studying the yields, y kg, from tomato plants. The data from a random sample of 70 tomato plants is summarised below.

Yield (y kg)	**Frequency** (f)	**Yield midpoint** (x kg)
$0 \leq y < 5$	16	2.5
$5 \leq y < 10$	24	7.5
$10 \leq y < 15$	14	12.5
$15 \leq y < 25$	12	20
$25 \leq y < 35$	4	30

(You may use $\Sigma fx = 755$ and $\Sigma fx^2 = 12037.5$)

Estimate the mean and the standard deviation of the yields of the tomato plants.

Mark scheme

$\bar{x} = 755/70 \approx 10.8$ kg (B1)

$\sigma_x = \sqrt{12037.5/70 - \bar{x}^2} = \sqrt{55.6326\ldots}$ (M1 A1ft)

$\sigma_x \approx 7.46$ kg. Accept $s \approx 7.51$ kg. (A1; 4 marks)

Calculating σ^2 From Combined Sets — Solution 72

Example: In classes 7A and 7B, consisting of 20 and 30 students respectively, the sums of their maths scores were $\Sigma a = 1200$ and $\Sigma b = 2100$, and $\Sigma a^2 = 75000$ and $\Sigma b^2 = 160000$.
Determine the standard deviation of maths scores in the two classes combined.

Let x be the scores across both classes.

The number of values, n, is just the total students across the two sets.

n $n = 20 + 30 = 50$

Similarly, the total score, Σx, across both classes is just the sum from each class:

Σx and Σx^2 $\Sigma x = \Sigma a + \Sigma b = 3300$

$\Sigma x^2 = \Sigma a^2 + \Sigma b^2 = 235000$

Once we have n, Σx and Σx^2 across the combined set, calculate the standard deviation:

σ

$\overline{x} = \frac{\Sigma x}{n} = \frac{3300}{50} = 66$

$\sigma^2 = \frac{\Sigma x^2}{n} - \overline{x}^2 = \frac{235000}{50} - 66^2 = 344$

$\sigma = \sqrt{344} = \mathbf{18.5}$ to 3sf

Further Example

Solution 73

In classes 8A and 8B, consisting of 5 and 10 students respectively, the means of their maths scores were 60 and 70, and the standard deviations 6 and 7.
Determine the standard deviation across all 15 students.

1. Use $\overline{x} = \frac{\Sigma x}{n}$ formula to determine Σx

Σa?

$\overline{a} = \frac{\Sigma a}{n}$

$60 = \frac{\Sigma a}{5} \quad \rightarrow \Sigma a = 300$

2. Similarly use σ^2 formula backwards to determine Σx^2

Σa²?

$\sigma_a{}^2 = \frac{\Sigma a^2}{n} - \overline{a}^2$

$6^2 = \frac{\Sigma a^2}{5} - 60^2 \quad \rightarrow \quad \Sigma a^2 = 18180$

3. Repeat the above steps for class 8B

Σb and Σb²?

$70 = \frac{\Sigma b}{10} \quad \rightarrow \quad \Sigma b = 700$

$7^2 = \frac{\Sigma b^2}{10} - 70^2 \quad \rightarrow \quad \Sigma b^2 = 49490$

4. combine sums together.

σ

$\Sigma x = 300 + 700 = 1000$

$\Sigma x^2 = 18180 + 49490 = 67670$

$n = 5 + 10 = 15$

$\overline{x} = \frac{1000}{15} = \frac{200}{3}$

$\sigma^2 = \frac{67670}{15} - \left(\frac{200}{3}\right)^2$

$= \frac{602}{9}$

$\sigma = \mathbf{8.179}$

Test Your Understanding

Solution 74

Exam Question

Throughout the year Alice took nine written tests and five practical tests in Music.

The mean mark of her nine written tests was 69.

The mean mark of her five practical tests was 83.

The standard deviation of her nine written tests was 15.4.

The standard deviation of her five practical tests was 7.8.

Calculate the standard deviation of all 14 tests.

(4 marks)

$69 = \frac{\Sigma w}{9} \quad \rightarrow \Sigma w = 621$

$15.4^2 = \frac{\Sigma w^2}{9} - 69^2 \quad \rightarrow \quad \Sigma w^2 = 44983.44$

$83 = \frac{\Sigma b}{5} \quad \rightarrow \quad \Sigma b = 415$

$7.8^2 = \frac{\Sigma b^2}{5} - 83^2 \quad \rightarrow \quad \Sigma b^2 = 34749.2$

$\Sigma x = 621 + 415 = 1036$

$\Sigma x^2 = 44983.44 + 34749.2$

$= 79732.64$

$n = 14$

$\overline{x} = \frac{1036}{14} = 74$

$\sigma^2 = \frac{79732.64}{14} - (74)^2$

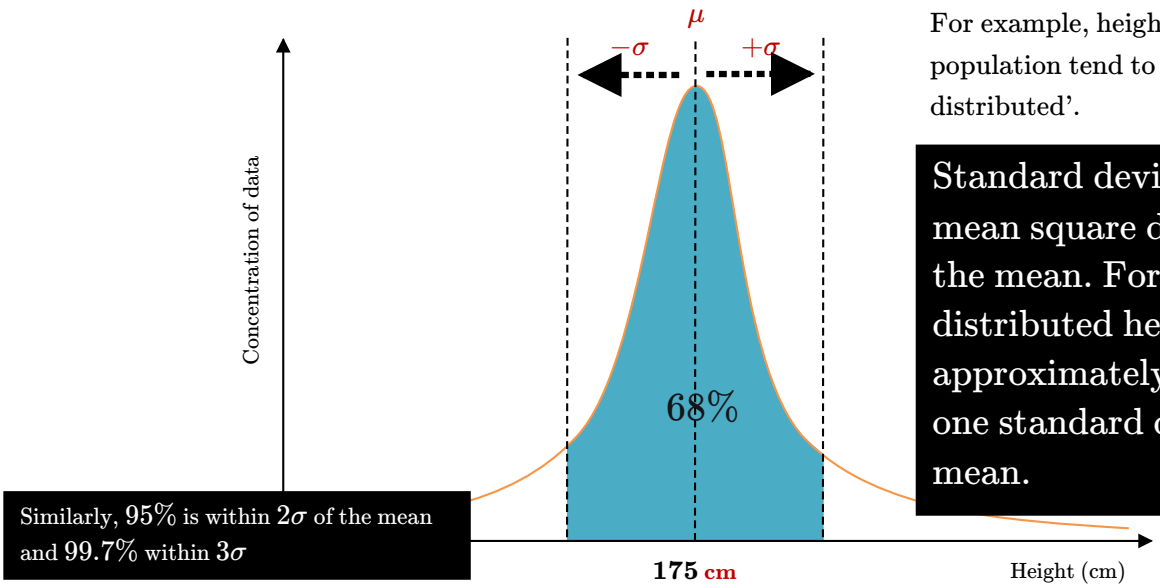
$= \frac{38358}{175}$

$\sigma = \mathbf{14.81}$

Looking Ahead

Solution 75

Standard deviation can be used as a measure of spread **for data with any 'shape'**. However, it becomes **particularly important when we consider normally distributed data**, i.e. has a symmetrical 'bell-curved' shape with data becoming less concentrated further away from the mean.



For example, heights in a population tend to be 'normally distributed'.

Standard deviation is the root mean square distance from the mean. For normally distributed heights, approximately 68% lie within one standard deviation of the mean.

Similarly, 95% is within 2σ of the mean and 99.7% within 3σ

You can find out about the normal distribution in skills 640-645.

Exercise 2E Page 32-33

Solution 76

2. Ten collie dogs are weighed. Use the summaries to find the standard deviation. (2 marks)

$$\sum w = 241, \qquad \sum w^2 = 5905$$

2

$\sigma = \sqrt{5905/10 - (241/10)^2} \approx 3.11\,\text{kg}$

4. Find the mean and standard deviation for the two sets combined.

Set	n	$\sum x$	$\sum x^2$
1	10	50	310
2	15	86	568

4

$\bar{x} = 136/25 = 5.44, \qquad \sigma = \sqrt{878/25 - 5.44^2} \approx 2.35$

5. Weekly pocket money was recorded to the nearest pound.

Pounds	8	9	10	11	12
f	14	8	28	15	20

(a) Use a calculator to find the mean and standard deviation. Give units. (3 marks)

(b) Estimate how many received more than one standard deviation above the mean. (2 marks)

5a

$\bar{x} = 869/85 \approx £10.22, \quad \sigma = \sqrt{9039/85 - \bar{x}^2} \approx £1.35$

5b

$\bar{x} + \sigma \approx £11.57, \qquad 20(12.50 - 11.57) \approx 19$ students

6. Calculate the standard deviation of days absent in one term. (2 marks)

Days absent	0	1	2	3	4
f	12	20	10	7	5

6

$\sigma = \sqrt{203/54 - (81/54)^2} \approx 1.23$ days

7. Lifetimes of 50 machine parts are grouped by hours (upper endpoint included).

Hours	5–10	10–15	15–20	20–25	25–30
f	5	14	23	6	2

Assess the claim that 90% lasted longer than one standard deviation below the mean.

7

$\bar{h} \approx 16.1, \quad \sigma \approx 4.69, \quad \bar{h} - \sigma \approx 11.41$

$50 - [5 + 14(11.41 - 10)/5] \approx 41$

About 82% exceeded the threshold. The data do not support the claim.

8. Daily mean windspeeds in Leeming, June 2015, ranged from 4 to 17 knots.

$n = 30, \quad \sum x = 243, \quad \sum x^2 = 2317$

(a) Use a calculator to find the mean and standard deviation. (2 marks)

(b) Estimate how many days exceeded the mean by more than one standard deviation. (2 marks)

(c) State one assumption used in your estimate. (1 mark)

8a

$\bar{x} = 8.1\,\text{kn}, \quad \sigma = \sqrt{2317/30 - 8.1^2} \approx 3.41\,\text{kn}$

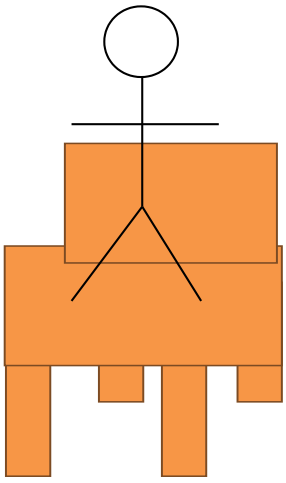
8b

$30(17 - 11.51)/(17 - 4) \approx 12.7 \text{ days} \approx 13 \text{ days}$

8c

Assume windspeeds are uniformly distributed over the range.

What do you reckon is the mean height of people in a room if they stand on a chair?



Is there an easy way to recalculate the mean based on the new heights? And the variance of the heights?

The mean would increase by the height of the chairs. The spread however is unaffected thus the variance would remain the same.

Starter

Solution 81

Suppose now after a bout of 'stretching you to your limits', you're now all 3 times your original height.

What do you think happens to the standard deviation of your heights?

It becomes 3 times larger (i.e. your heights are 3 times as spread out!)

What do you think happens to the variance of your heights?

It becomes 9 times larger. We use the scale factor of the standard deviation, squared.

Extension Question: Can you prove the latter using the formula for variance?

$$\sigma^2 = \frac{\Sigma(3x)^2}{n} - \left(\frac{\Sigma(3x)}{n}\right)^2$$
$$= \frac{\Sigma 9x^2}{n} - \left(\frac{3\Sigma x}{n}\right)^2 = \frac{9\cdot\Sigma x^2}{n} - 9\left(\frac{\Sigma x}{n}\right)^2$$
$$= 9\left(\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2\right)$$

Rules of coding

Solution 82

Suppose our original variable (e.g. heights in cm) was x.
Then y would represent the heights with 10cm added on to each value.

Coding	Effect on $\overline{x}$	Effect on σ
$y = x + 10$	$\overline{x}$ will similarly increase by 10 (to get $\overline{y}$)	As discussed, adding (and subtracting) has no effect on standard deviation or any measure of spread.
$y = 3x$	$\overline{x}$ will get 3 times bigger.	Standard deviation will get 3 times larger.
$y = 2x - 5$	$\overline{y} = 2\overline{x} - 5$, i.e. effect on values is same effect on mean.	-5 has no effect but standard deviation will get 2 times larger.

You might get any **linear** coding (i.e. using $\times + \div -$). We might think that any operation on the values has the same effect on the mean. But note for example that **squaring** the values would not square the mean; we already know that $\Sigma x^2 \neq (\Sigma x)^2$ in general.

The point of coding

Solution 83

Cost x of diamond ring (£)

£1010 £1020 £1030 £1040 £1050

We 'code' our variable using the following:

$$y = \frac{x-1000}{10}$$

New values y:

£1 £2 £3 £4 £5

Standard deviation of y, σ_y: $\sqrt{2}$

Therefore...

Standard deviation of x, σ_x: $10\sqrt{2}$

The jist of coding: We want to find the mean/standard deviation of a variable. We transform the values, using some rule, to make them simpler. We can then more easily calculate the mean/standard deviation of the 'coded' data, and from this we can then determine what the mean/standard deviation would have been for the original uncoded data.

Quickfire questions

Solution 84

Old mean $\overline{x}$	Old σ_x	Coding	New mean $\overline{y}$	New σ_y
36	4	$y = x - 20$	16	4
36	8	$y = 2x$	72	16
35	4	$y = 3x - 20$	85	12
40	3	$y = \frac{x}{2}$	20	$\frac{3}{2}$
11	27	$y = \frac{x+10}{3}$	7	9
300	25	$y = \frac{x-100}{5}$	40	5

Example Exam question

Solution 85

The table summarises the times, to the nearest minute, for a group of students to complete an exam.

Time (min)	11–20	21–25	26–30	31–35	36–45	46–60
Students	62	88	16	13	11	10

You may use $\sum ft^2 = 134281.25$.

(a) Estimate the mean and standard deviation. (5)

(b) Use linear interpolation to estimate the median. (2)

(c) Show that the estimated lower quartile is 18.6 to 3 significant figures. (1)

(d) Estimate the interquartile range. (2)

(e) Give a reason why the mean and standard deviation are not the most appropriate summary statistics. (1)

Suppose we have worked all these out already.

The timekeeper made an error. Each student took 5 minutes less than recorded. The corrected times are shown below.

Time (min)	6–15	16–20	21–25	26–30	31–40	41–55
Students	62	88	16	13	11	10

(f) Without further calculations, explain the effect on the estimates in (a), (b), (c) and (d). (3)

f

Coding: $t_2 = t_1 - 5$.

The mean, median and lower quartile each decrease by 5 minutes.

The standard deviation and interquartile range are unchanged.

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Solution 86

3. Television prices are coded as follows. Find the original mean price. (2 marks)

$y = (x - 65)/200, \qquad \bar{y} = 1.5$

3

$\bar{x} = 200(1.5) + 65 = 365$

6. Weekly incomes of 100 workers are coded. Find the standard deviation of their original incomes. (2 marks)

$y = (i - 90)/100, \quad \sum y = 131, \quad \sum y^2 = 176.84$

6

$\sigma_i = 100\sqrt{176.84/100 - (131/100)^2} \approx 22.9$

7. Annual rainfall at six places is coded. Estimate the standard deviation in millimetres. (2 marks)

$s = 0.01x - 10, \quad \sum s = 16.1, \quad \sum s^2 = 147.03$

7

$\sigma_x = 100\sqrt{147.03/6 - (16.1/6)^2} \approx 416\,\text{mm}$

9. Pressures in Hurn in June 2015 are coded. Find the original mean and standard deviation. (4 marks)

$c = p/2 - 500, \quad n = 30, \quad \bar{c} = 10.15, \quad S_{cc} = 296.4$

9

$\bar{p} = 2(10.15 + 500) = 1020.3\,\text{hPa}, \qquad \sigma_p = 2\sqrt{296.4/30} \approx 6.29\,\text{hPa}$

8. A teacher adds 12 to each mark and then reduces the result by 20%.

(a) Write a formula for the standardised mark t in terms of the original mark m. (1 mark)

(b) Find the original mean and standard deviation given the following summaries. (3 marks)

$n = 28, \qquad \bar{t} = 52.8, \qquad S_{tt} = 7.3$

8a

$t = 0.8(m + 12)$

8b

$\bar{m} = 52.8/0.8 - 12 = 54$

$\sigma_m = \frac{1}{0.8}\sqrt{7.3/28} \approx 0.64$

I have a list of 30 heights in the class. What item do I use for:

Q1 8th item

Q2 Between the 15th and 16th items

Q3 23rd item

For the following grouped frequency table, calculate:

Height **h** of bear (in metres)	Frequency
$0 \leq h < 0.5$	4
$0.5 \leq h < 1.2$	20
$1.2 \leq h < 1.5$	5
$1.5 \leq h < 2.5$	11

a) The estimate mean: $\overline{h} = \frac{(0.25\times4)+(0.85\times20)+\ldots}{40} = \frac{46.75}{40} = 1.17m to 3sf$

b) The estimate median: $0.5 + \left(\frac{16}{20} \times 0.7\right) = 1.06m$

c) The estimate variance:
(you're given $\Sigma f h^2 = 67.8125$)

$\sigma^2 = \frac{67.8125}{40} - \left(\frac{46.75}{40}\right)^2 = 0.329\,\text{m}^2 to 3sf$

What is the standard deviation of the following lengths: 1cm, 2cm, 3cm

$\sigma^2 = \frac{14}{3} - 2^2 = \frac{2}{3}$ $\sigma = \sqrt{\frac{2}{3}}\,\text{cm}$

The mean of a variable x is 11 and the variance 4.
The variable is coded using $y = \frac{x+10}{3}$. What is:

a) The mean of y? a) $\overline{y} = 7$

b) The variance of y? b) $\sigma_y^2 = \frac{4}{9}$

A variable x is coded using $y = 4x - 5$.
For this new variable y, the mean is 15 and the standard deviation 8.
What is:

c) The mean of the original data? c) $\overline{x} = 5$

d) The standard deviation of the original data? d) $\sigma_x = 2$